

KHMELNYSKYI, Ukraine

WMO: 334290

Lat: 49.3536N

Lon: 26.9375E

Elev: 350

StdP: 97.19

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.6	-20.5	0.6	-17.3	-17.4	0.8	-14.4	12.0	-0.4	10.7	-1.1	3.3	60	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.6	20.0	27.9	19.2	26.4	18.5	21.3	27.4	20.4	26.0	19.5	24.6	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.6	24.0	18.4	13.8	23.0	17.6	13.1	22.2	63.4	27.5	60.1	25.9	57.1	24.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	9.0	8.0	-20.8	31.9	3.8	1.4	-23.6	33.0	-25.8	33.8	-27.9	34.6	-30.6	35.7
			-21.1	23.1	3.7	1.6	-23.7	24.3	-25.9	25.2	-28.0	26.2	-30.7	27.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-4.2	-2.7	1.9	9.1	14.5	17.7	19.5	19.1	14.1	8.2	2.8	-2.4
	DBStd	9.55	5.40	5.49	4.74	4.35	3.95	3.34	3.07	3.26	3.88	4.28	4.67	5.14
	HDD10.0	1820	441	356	253	66	8	0	0	0	9	86	218	384
	HDD18.3	3866	699	589	509	277	129	52	23	30	136	313	466	643
	CDD10.0	1161	0	0	1	39	148	230	296	281	132	31	3	0
	CDD18.3	165	0	0	0	1	11	32	61	53	8	0	0	0
	CDH23.3	1289	0	0	0	7	93	223	457	435	74	1	0	0
	CDH26.7	292	0	0	0	1	10	33	113	122	13	0	0	0
(14) Wind	WSAvg	4.0	4.5	4.6	4.7	4.3	3.9	3.6	3.4	3.3	3.6	3.9	4.4	4.3
(15) Precipitation	PrecAvg	641	33	35	36	47	55	89	99	67	60	42	42	38
	PrecMax	878	59	61	88	141	116	188	246	209	157	138	86	98
	PrecMin	410	7	10	9	10	18	33	32	1	8	1	3	14
	PrecStd	107	13	14	21	26	24	39	58	46	41	28	19	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	10.1	16.1	23.6	28.1	29.3	31.7	32.1	28.4	22.3	15.7	9.2
		MCWB	5.2	6.7	9.3	14.1	17.9	20.0	21.2	20.1	18.0	15.1	11.6	7.8
	2%	DB	4.8	7.1	13.0	20.7	25.9	27.6	29.6	29.7	25.4	19.2	12.5	7.1
		MCWB	3.6	5.2	7.9	12.6	16.5	19.3	20.5	19.8	16.4	13.8	10.2	6.0
	5%	DB	3.4	5.2	10.6	18.4	23.8	26.0	27.7	27.8	23.2	16.5	10.7	5.1
		MCWB	2.4	3.8	6.5	11.4	16.1	18.6	19.8	19.0	15.6	12.5	9.1	4.3
	10%	DB	2.0	3.5	8.3	16.3	21.7	24.3	25.9	25.7	20.7	14.3	9.1	3.5
		MCWB	1.2	2.3	5.3	10.2	14.9	17.9	19.1	18.3	14.8	11.3	7.9	2.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	7.3	10.2	14.9	19.5	22.0	22.5	22.4	19.2	16.3	12.1	8.2
		MCDB	6.6	9.6	14.9	22.5	25.4	27.3	28.5	29.0	26.0	20.4	14.4	9.0
	2%	WB	3.8	5.4	8.4	13.4	17.8	20.5	21.4	21.0	17.5	14.3	10.7	6.0
		MCDB	4.6	6.9	12.1	19.1	23.6	25.6	27.7	27.4	23.0	17.8	12.3	6.9
	5%	WB	2.5	3.9	6.9	12.2	16.7	19.5	20.5	19.9	16.4	13.0	9.3	4.4
		MCDB	3.2	5.1	9.6	17.2	22.2	24.5	26.3	25.3	21.3	16.1	10.5	5.1
	10%	WB	1.3	2.5	5.6	10.9	15.6	18.5	19.6	19.0	15.4	11.6	7.9	2.8
		MCDB	1.9	3.5	8.0	15.2	20.6	23.1	24.7	24.2	19.7	13.9	9.1	3.4
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.0	6.5	8.7	9.6	9.1	9.4	9.9	8.6	7.1	4.7	4.2
		MCDBR	4.5	6.8	10.3	11.9	12.4	11.5	12.1	12.7	12.5	10.6	7.8	5.2
		MCWBR	3.6	5.0	6.2	5.6	5.6	5.3	4.9	5.0	5.4	5.7	5.2	4.3
	5% WB	MCDWR	4.2	6.3	9.3	10.8	11.0	10.2	10.7	11.2	10.6	9.3	6.9	4.9
		MCWBR	3.6	4.9	6.0	5.5	5.4	5.2	4.9	5.0	5.4	5.4	5.0	4.2
(40) Clear-Sky Solar Irradiance	taub		0.306	0.337	0.375	0.429	0.403	0.402	0.433	0.432	0.393	0.360	0.337	0.304
	taud		2.237	2.190	2.200	2.124	2.270	2.291	2.219	2.215	2.284	2.359	2.336	2.305
	Ebn at Noon		725	789	824	814	857	858	823	803	797	761	687	678
	Edh at Noon		75	100	118	142	128	127	134	129	108	85	69	62
(44) All-Sky Solar Radiation	RadAvg		0.95	1.79	2.97	4.33	5.55	5.88	5.64	4.97	3.48	2.13	0.92	0.66
	RadStd		0.18	0.28	0.41	0.46	0.48	0.44	0.41	0.36	0.53	0.31	0.10	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.74	N/A	N/A	+0.91	+0.51	N/A	N/A	-190	+137	+53
(47)	Regional (13 neighbors)	+0.70	N/A	N/A	+0.95	+0.50	+0.44	-139	-210	+126	+52

Nomenclature: See separate page